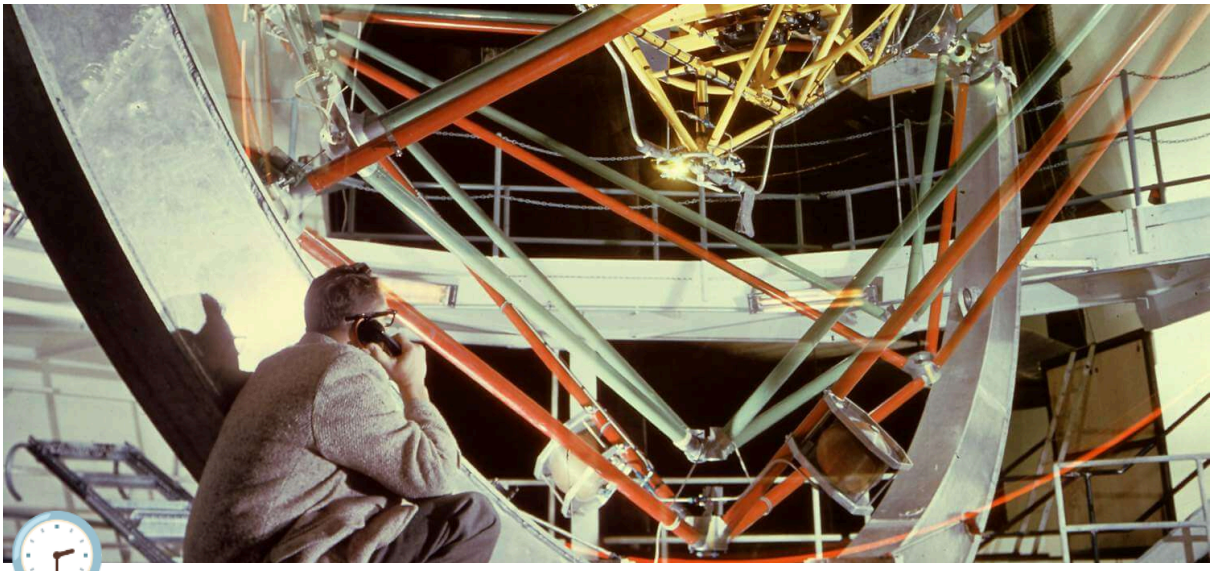


2022 Paper

Question 1

Implement the following Boolean function with 4 x 1 multiplexer and external gates

$$F(A,B,C,D) = \sum(1,2,5,7,8,10,11,13,15)$$

Connect the inputs A and B to the selection lines. The input requirements for the four data lines will be a function of variables C and D. These values are obtained by expressing F as a function of C and D for each of the four cases when AB = 00, 01, 10 and 11. These functions may have to be implemented with external gates.

Draw a circuit diagram of your solution and provide a truth table.

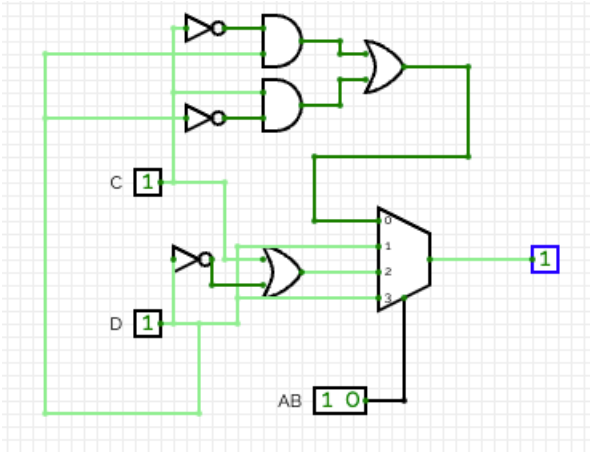
Attempt:

Change this to product of sum

$$\sum(1,2,5,7,8,10,11,13,15) = \Pi(0,3,4,6,9,12,14)$$

Draw K-map:

	AB	00	01	11	10
CD	-	-	-	-	-
00	|	1	1	1	
01	|				1
11	|	1			
10	|		1	1	



The truth table is just the min terms.

Question 2

Please notice, the following processor has a Function Unit that implements for the (FS = 00011) vector a multiplication (F = A * B) operation instead of a (F = A + B + 1) operation (see Figure 6). Furthermore, the processor has 7 “Temp Registers” that may be selected from the control memory (see Figure 4 and Figure 5).

Would it be possible to implement a 3x3 Matrix Multiplication Instruction **C**=**AB** in the control memory?

Are the 7 “Temp Registers” sufficient to implement all the matrix multiplication indices, pointers, and other storage necessary?

Assuming the memory holds three 3x3 matrices **A**, **B**, and **C** in row-major order and the 3x3 Matrix Multiplication Instruction provides addresses for two registers in the register-file, that hold pointers to the first elements **a11** and **b11** of these two matrices **A** and **B** (see Figure 1). Element **c11** of the matrix product **C** follows the address of the last element **b33** of the matrix **B.**

Design an Algorithm State Machines (ASM) for a 3x3 Matrix Multiplication Instruction that can be implemented on a modified version of the processor that you designed and simulated as part of your coursework for the "CSU22022 - Computer Architecture I" module. The processor architecture is shown in Figure 4. If the 7 “Temp Registers” are insufficient, change the hardware design to a 15 “Temp Registers” solution. Consequently, TD, TA, and TB would be 4 bit vectors. This needs to be considered in the control memory template Figure 5.

a) You should provide a ASM chart for the 3x3 Matrix Multiplication Instruction and a written description for the various states and the state transitions of the ASM. This should include all register transfer operations that need to be executed.

b) You should also populate the first 10 consecutive Control Memory Addresses that implement the first 10 states of your ASM. Therefore, showing how your ASM design is implemented on the modified architecture Figure 4. A Control Memory template is shown in Figure 5.

Please assume the following: The 3x3 Matrix Multiplication Instruction has the following format Figure 1:

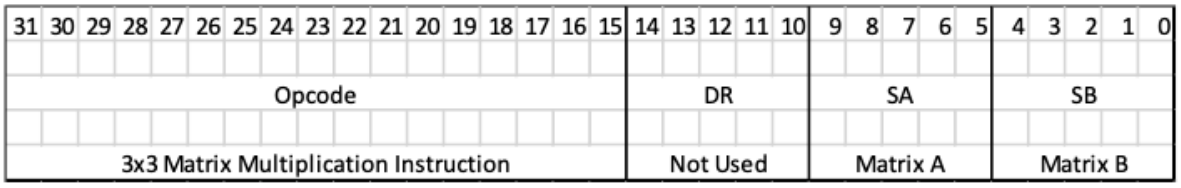


Figure 1 3x3 Matrix Multiplication Instruction

SA – is the address of the register that holds a pointer to the first elements a11 of the **A** matrix. The **A** matrix elements provide the input for the 3x3 matrix multiplication.

SB – is the address of the register that holds a pointer to the first elements b11 of the **B** matrix. The **B** matrix elements provide the input for the 3x3 matrix multiplication.

The processor Figure 4 implements the functions shown in Figure 2. The F = srB and F slB functions perform a single bit shift. They insert a '0' in the vacated MSB or LSB position. Furthermore, these shift operations will not set the carry flag.

Table 1: FS code definition

FS	MF Select	G Select	H Select	Micro-operation
00000	0	0000	00	$F = A$
00001	0	0001	00	$F = A + 1$
00010	0	0010	00	$F = A + B$
00011	0	0011	00	**F = A * B**
00100	0	0100	01	$F = A + \bar{B}$
00101	0	0101	01	$F = A + \bar{B} + 1$
00110	0	0110	01	$F = A - 1$
00111	0	0111	01	$F = A$
01000	0	1000	00	$F = A \wedge B$
01010	0	1010	10	$F = A \vee B$
01100	0	1100	10	$F = A \oplus B$
01110	0	1110	10	$F = \bar{A}$
10000	1	0000	00	$F = B$
10100	1	0100	01	$F = srB$
11000	1	1000	10	$F = slB$

Figure 2 Function Code

The modified processor shown in Figure 4 has changed the T-signals to 3 bit TD, 3 bit TA, and 3 bit TB signals. This provides you with 7 "Temp Registers". Figure 3 shows how you can access these registers. If TD, TA, or TB are set to '000' DR, SA, and SB will select one of the 32 registers in the register-file. Settings other than '000' will give you access to the 7 "Temp Registers".

TD				TA				TB			
0	0	0	DR selected Register	0	0	0	SA selected Register	0	0	0	SB selected Register
0	0	1	T0	0	0	1	T0	0	0	1	T0
0	1	0	T1	0	1	0	T1	0	1	0	T1
0	1	1	T2	0	1	1	T2	0	1	1	T2
1	0	0	T3	1	0	0	T3	1	0	0	T3
1	0	1	T4	1	0	1	T4	1	0	1	T4
1	1	0	T5	1	1	0	T5	1	1	0	T5
1	1	1	T6	1	1	1	T6	1	1	1	T6

Figure 3 TD-DR, TA-SA, and TB-SB

Figure 5 Control Memory with 3 bit TD, 3 bit TA, and 3 bit TB




```
architecture Behavioral of control_memory is
type mem_array is array(0 to 255) of std_logic_vector(47 downto 0);

begin

memory_m: process(IN_CAR)
        variable control_mem : mem_array:=(
-- |47           31|3028|27|26|25|24|2321|2018|1715|14|13  9|8|7|6|5|4|3|2|1|0|
-- | Next Address  | MS | M| I| P| P| TD | TA | TB | M|  FS |M|R|M|M|R|R|R|R|F|
-- |               |    | C| L| I| L|    |    |    | B|     |D|W|M|W|V|C|N|Z|L|
  "00000000000000000 000  0  0  0  0  000  000  000  0 00000 0 0 0 0 0 0 0 0 0",-- 00
  "00000000000000000 000  0  0  0  0  000  000  000  0 00000 0 0 0 0 0 0 0 0 0",-- 01
  "00000000000000000 000  0  0  0  0  000  000  000  0 00000 0 0 0 0 0 0 0 0 0",-- 02
  "00000000000000000 000  0  0  0  0  000  000  000  0 00000 0 0 0 0 0 0 0 0 0",-- 03
  "00000000000000000 000  0  0  0  0  000  000  000  0 00000 0 0 0 0 0 0 0 0 0",-- 04
  "00000000000000000 000  0  0  0  0  000  000  000  0 00000 0 0 0 0 0 0 0 0 0",-- 05
  "00000000000000000 000  0  0  0  0  000  000  000  0 00000 0 0 0 0 0 0 0 0 0",-- 06
  "00000000000000000 000  0  0  0  0  000  000  000  0 00000 0 0 0 0 0 0 0 0 0",-- 07
```